DATA ANALYSIS AND PROGRAMMING FOR OPERATIONS MANAGEMENT

Final Report

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Stock Selection Decision Support Tool for Shooting Star Capital

Introduction

Shooting Star Capital is currently setting up a new equity fund and aims to support the stock selection process with a structured data analysis workflow.
The objective of this project is to develop a decision support tool that automates the management of market data, analyzes sector performance, and determines an **optimal portfolio composition** under predefined investment rules.

The dataset consists of **500 stocks** collected in individual text files. Each file contains the stock identifier, the industry sector of the respective company, the number of outstanding shares, and the daily closing prices for the past year.

The work carried out in this project follows four main steps:

1. Organizing the raw data into a structured SQLite **database**.
2. Analyzing **sector performance** in 2025 through market-capitalization-weighted returns.
3. Designing an **optimal portfolio allocation** using mathematical optimization.
4. Evaluating the sensitivity of the resulting portfolio to changes in the investment strategy.

1. Data Organization and Database Structure

All market data was transferred into a structured SQLite relational database, which organizes the information into two main components. The first contains general information about each **stock**, including the stock identifier, the industry sector, and the number of outstanding shares. The second contains historical **price** observations, storing the closing price of each stock for every trading day.

This structure ensures that the data is stored in a consistent format and can be efficiently retrieved through database queries. It also enables flexible filtering mechanisms that allow users to retrieve only relevant stocks.

In particular, the decision support tool allows the user to **filter stocks based on two criteria**: the industry sector and the last known closing price. This functionality allows analysts to quickly restrict the analysis to specific segments of the market or to stocks that satisfy a given price threshold.

2. Sector Performance Analysis

Since no prior management directive specifies a focus on particular sectors, the analysis considers the **entire market**. The first objective is therefore to obtain an **overview of sector performance during 2025**.

For each stock, the annual return is computed using the first and last price observations available in the dataset. The first price is interpreted as the last closing price of 2024 and therefore represents the reference value for computing the return in 2025: This measure represents the stock performance over the year.

To better understand the characteristics of the dataset, the distribution of **individual stock returns is examined**. This analysis provides insight into the variability of stock performance and helps identify the overall spread of returns across the market.

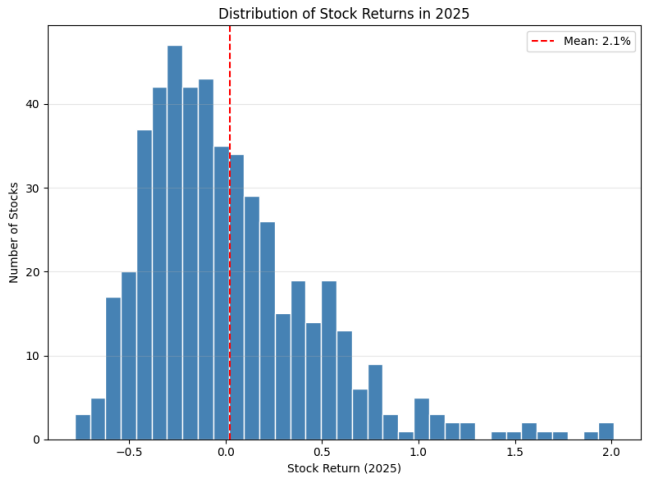

Figure 1 – Distribution of individual stock returns in 2025

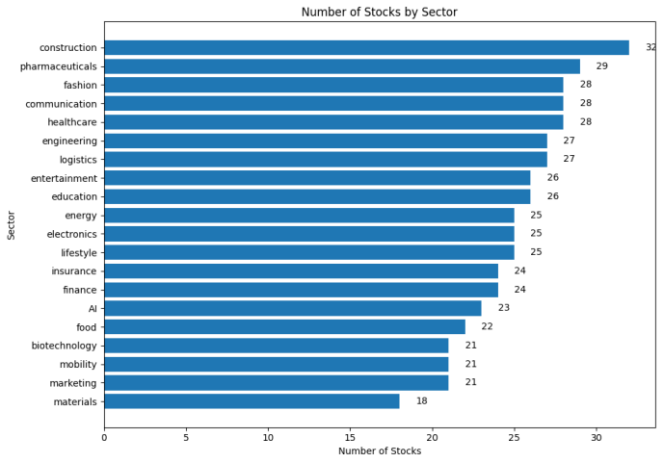

Figure 2 – Number of companies per industry sector

Sector performance is then evaluated by computing the average return of each sector. Instead of a simple arithmetic average, the analysis uses a m**arket-capitalization-weighted return**. Market capitalization is calculated as the product of the number of outstanding shares and the stock price. Weighting returns by market capitalization ensures that larger firms have a proportionally larger impact on the sector performance measure.

The results are visualized to highlight differences in performance across sectors.

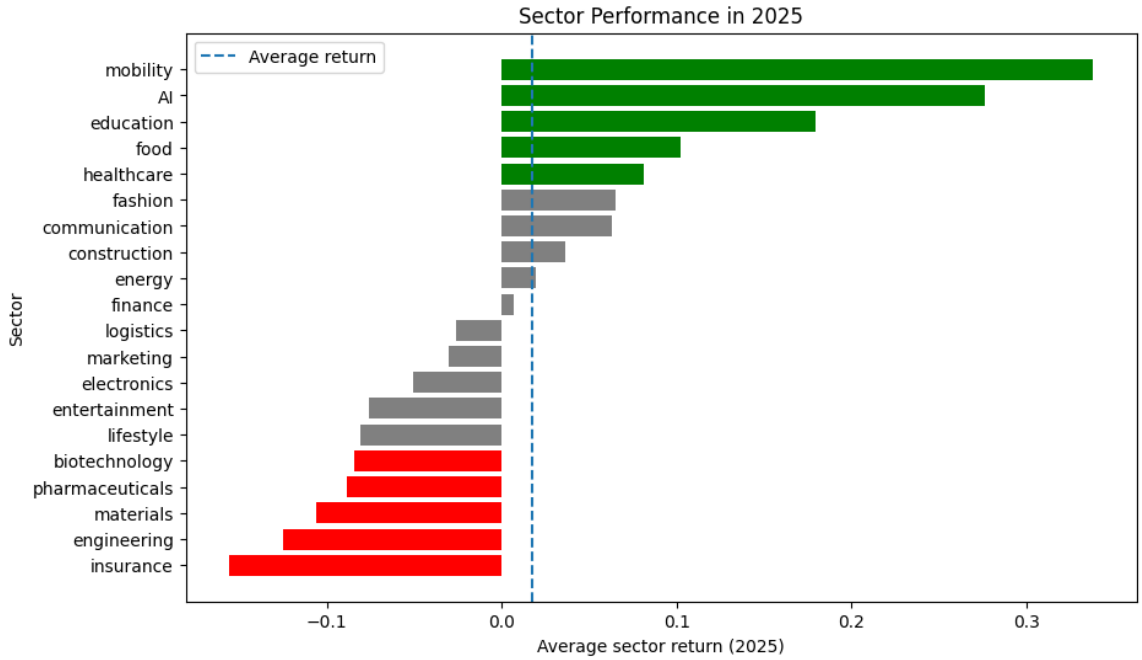

Figure 3 – Market-capitalization weighted average return per sector

Based on their performance, sectors are grouped into **three categories**:

- Top 25% performing sectors
- Middle 50% sectors
- Bottom 25% performing sectors

This classification forms the basis for the portfolio allocation strategy described in the next section.

3. Portfolio Optimization

After analyzing the sector performance, the next step is to determine the **composition of the equity fund**.

The portfolio design is based on the assumption that the relative performance observed in 2025 provides an indication of near-term future performance. Based on this assumption, **capital is allocated across sectors according to a predefined investment strategy**.

The investment rules are as follows:

- 50% of the available capital is invested in stocks belonging to the top **25% performing sectors.**
- 30% of the capital is invested in stocks from the middle **50% sectors.**
- 20% of the capital is invested in stocks from the bottom **25% sectors.**

In addition, to prevent excessive concentration risk, **no more than 5% of the total capital may be invested in a single company.**

The portfolio selection problem is formulated as a **mathematical optimization model**. The decision variables represent the fraction of capital invested in each stock. **The objective is to maximize the expected portfolio return** subject to the investment constraints described above.

Solving this model produces the optimal allocation of capital across the available stocks. Only the stocks receiving a positive allocation are included in the final portfolio. The resulting allocation identifies the stocks included in the portfolio and their investment weights.

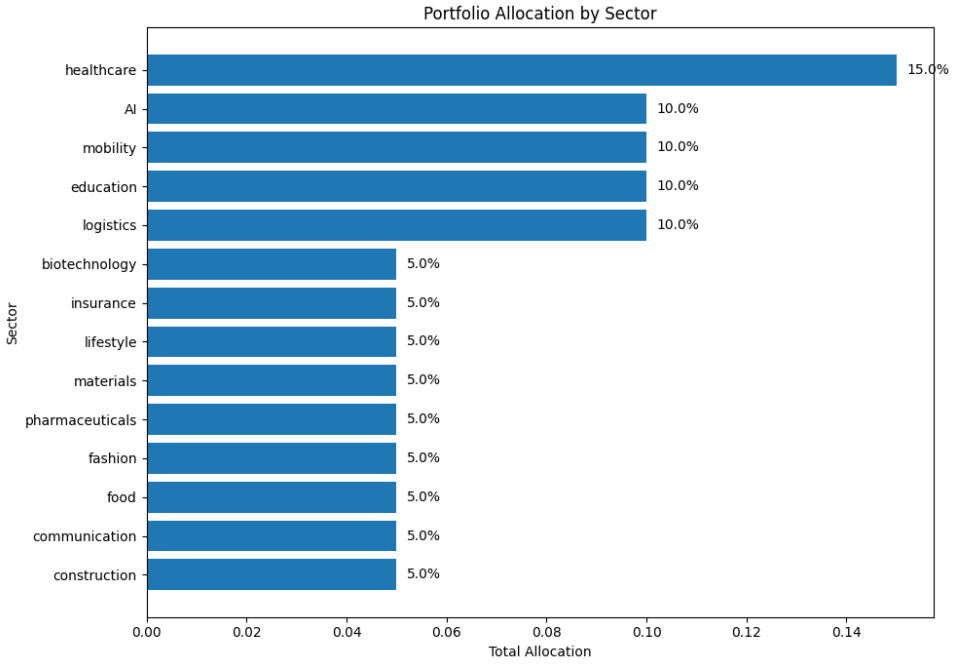

Figure 4 – Optimal portfolio allocation across selected stocks

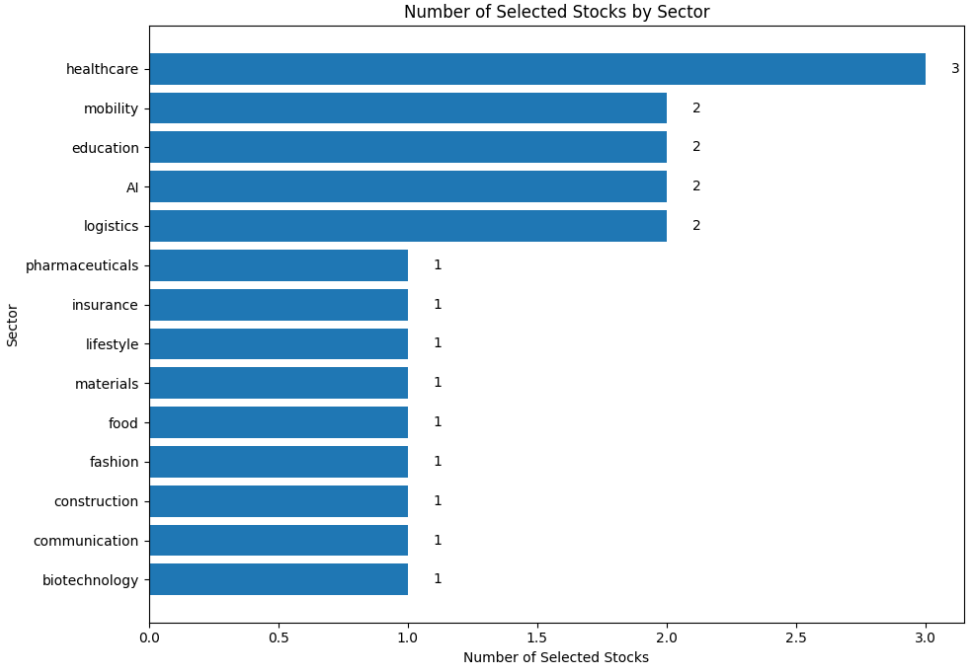

Figure 5 – Number of selected stocks by sector

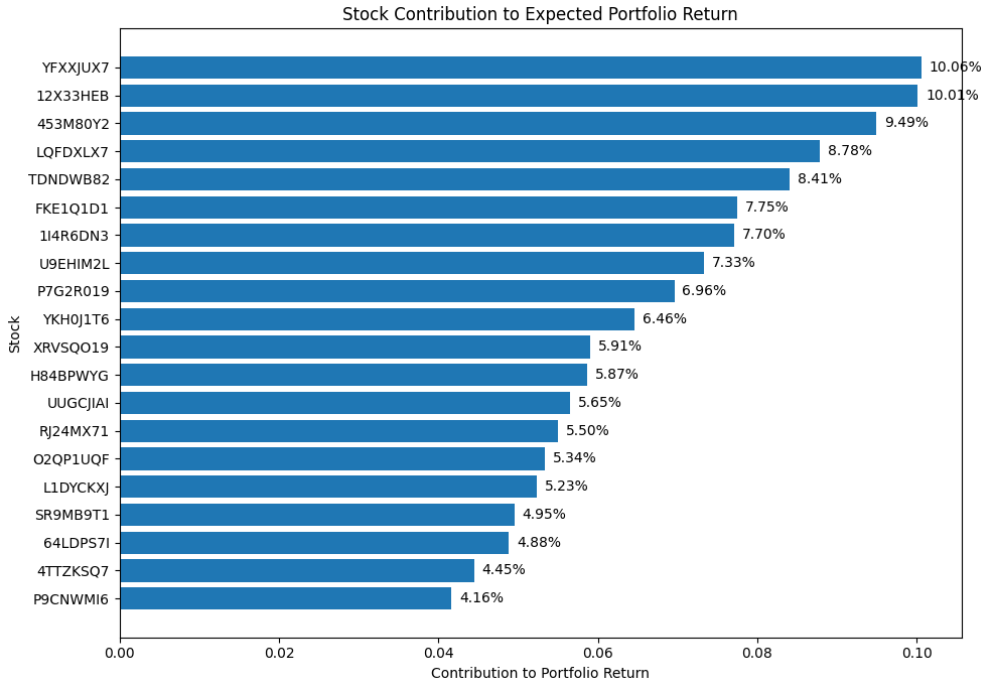

Figure 6 – Stock contribution to Expected Portfolio Return

4. Sensitivity Analysis

Portfolio performance can be sensitive to the investment constraints imposed by the strategy. To **assess the robustness of the results, a sensitivity analysis is performed**.

In particular, the effect of the **maximum investment allowed per company is evaluated**. While the baseline strategy restricts investment in a single company to 5% of the total capital, alternative scenarios are considered in which this limit is increased.

Three scenarios are analyzed:

- Maximum investment per stock equal to 5%
- Maximum investment per stock equal to 10%
- Maximum investment per stock equal to 20%

For each scenario, **the optimization model is solved again** and the expected portfolio return is recorded. **Comparing the resulting returns** allows an assessment of how strongly portfolio performance depends on the diversification constraint.

This analysis shows how the concentration constraint influences the achievable portfolio return.

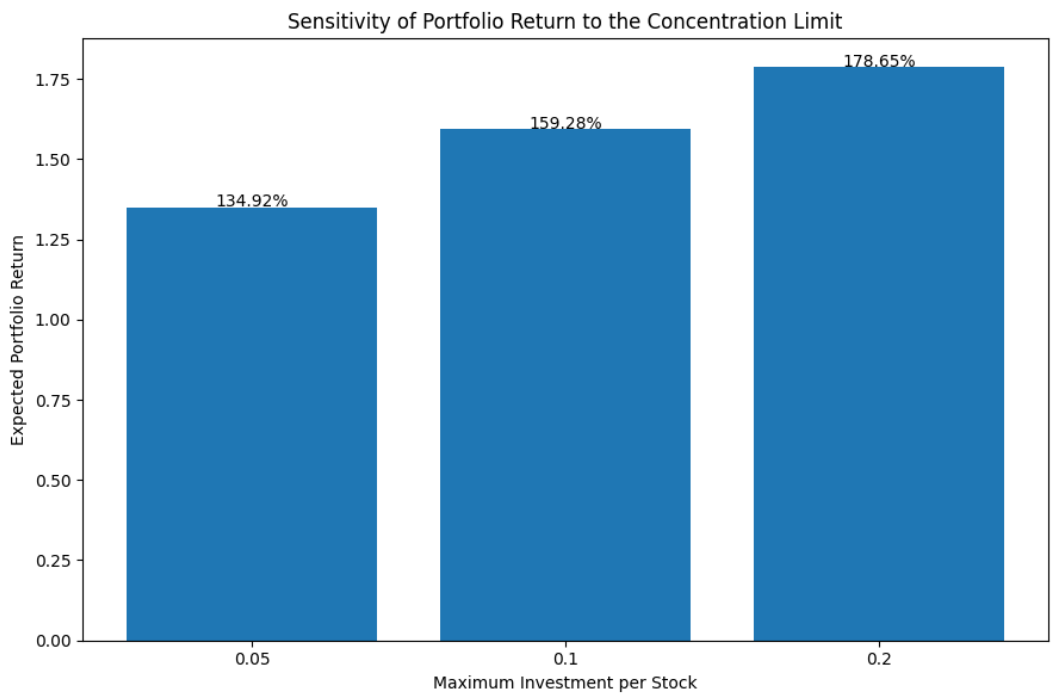

Figure 7 – Impact of concentration limits on expected portfolio return

Conclusion

This analysis provides a data-driven framework to support stock selection and portfolio construction for Shooting Star Capital.

The distribution of stock returns shows substantial variability across firms, with an average market return of approximately **2.1%**. While many stocks achieve moderate returns, a smaller group of companies produces significantly higher gains. This dispersion in performance highlights the importance of systematic stock selection when constructing a portfolio.

The sector analysis reveals strong differences in performance across industries. In particular, **mobility, AI and education** emerge as the best performing sectors, with average returns of roughly **34%, 28% and 18%**, respectively. In contrast, sectors such as **insurance, engineering and materials** show negative average returns, indicating weaker overall performance. These differences provide the basis for the sector classification used in the portfolio allocation strategy.

The optimization model allocates capital primarily to sectors with stronger performance while respecting the diversification constraint of **5% per stock**. As a result, the portfolio concentrates investments in sectors such as **healthcare (15%)**, **AI, mobility, education and logistics (around 10% each)**. At the same time, smaller allocations are distributed across additional sectors to maintain diversification.

The analysis of stock contributions shows that portfolio performance is mainly driven by a limited number of high-return stocks, which together account for a large share of the expected portfolio return. This confirms that the optimization model successfully prioritizes the most promising investment opportunities while respecting the imposed constraints.

Finally, the sensitivity analysis demonstrates that portfolio performance is strongly influenced by the concentration limit. When the maximum investment per stock increases from **5% to 10% and 20%**, the expected portfolio return rises from approximately **135% to 159% and 179%**, respectively. This result highlights the trade-off between diversification and return concentration: allowing larger positions increases the achievable return but also reduces portfolio diversification.

Overall, the decision support tool provides a structured approach for transforming market data into portfolio allocation decisions while clearly illustrating how sector performance and investment constraints influence the final portfolio.

Appendix

- https://chatgpt.com/share/69b85268-3440-800f-9b33-0f19e7fd60b9
- https://chatgpt.com/share/69b85785-d0e0-800f-aeda-3dfd4efac3fd
- https://chatgpt.com/share/69b8592b-b8f8-800f-bead-7f69d4ea5236
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- https://matplotlib.org/stable/gallery/index.html
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